

NGUYEN HONG DUY

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[MyPortfolio](#) | github.com/ChimUng

OBJECTIVE

Four-year Software Engineering student at Sai Gon University with a strong foundation in Java, JavaScript. Aiming to grow as a Full-stack Engineer with a long-term goal of building scalable, high-performance distributed systems. Passionate about system design, and performance optimization.

EDUCATION

Sai Gon University
Bachelor of Software Engineering
• GPA: 3.2/4.0

Expected Aug 2027
Ho Chi Minh City, Vietnam

SKILLS

Programming Languages (Proficient): Java, JavaScript, TypeScript

Framework (Proficient): Spring Boot, React, Next.js, Node.js

Databases (Proficient): MongoDB, PostgreSQL, Redis, Kafka

Cloud/Tools (Familiar): AWS, Cloudflare, Docker, Git, Postman, GitHub CI/CD

Languages: Vietnamese (Native), English (Advanced)

PROJECTS

Animeflix – Anime Streaming Platform  **ANIMEFLIX** Jun 2025 – Present
Next.js 15, Node.js, TypeScript, MongoDB, Redis, NextAuth, Vite, Cloudflare, SW, Pusher

- Built multi-tier cache (SW/PWA → region-aware Edge CDN → Redis → Origin), isolating invocation per-region; Redis caching cut response time 75–85%, eliminated 99% rate-limit errors.
- Designed Cursor Pagination (indexed, fixed page-size) + MongoDB document-clustered schema for comments/replies/reactions, avoiding OFFSET re-scan under concurrent load.
- Applied RESTful GET/POST semantics with tiered Cache-Control (public vs private/short-TTL) + ISR revalidation, limiting Origin hits to 1 per cycle regardless of traffic spikes.
- Architected SSR/ISR static-eligible pages with client-side dynamic fetch to preserve Edge cacheability; integrated Pusher WebSocket for real-time comment/notification delivery without polling overhead.
- Implemented HLS streaming with Edge-cached .ts segments, secured via server-to-server signed-URL token (short-TTL) to prevent hotlinking beyond browser-level CORS.

Animeflix Third-Party System [\[GitHub\]](#) Oct 2025 – Present
Spring Boot, Docker, MongoDB, Kafka, Redis, RAG, AWS

- Built multi-tier REST API cache (JVM Heap via Caffeine → Redis → Cloudflare Edge) for public GET endpoints, reducing round-trips to Origin for shared/non-user-specific data.
- Implemented stateless JWT auth (access/refresh rotation, 15-min/7-day TTL), eliminating per-request DB/session-store round-trip in favor of lightweight local signature verification.
- Designed event-driven microservices (catalog-service, ai-search-service) decoupled via Kafka, isolating heavy cron/AI workloads from user-facing search to prevent RAM exhaustion under concurrent load.
- Built Hybrid Search (Gemini 768-dim embeddings + MongoDB Atlas Vector Search) with confidence-based routing between structured keyword and semantic paths, tuned similarity threshold (0.78) and score-gap detection (0.025) across 5,000+ anime titles.
- Refactored search pipeline to reactive non-blocking (WebFlux, Mono.zip), cutting response latency from 2s to 600ms; added Circuit Breaker on Gemini API (429 handling) with fallback to semantic-only path.

ACTIVITIES & INTERESTS

- Familiar with Git workflows, pull requests, Postman API testing, and RESTful service debugging.
- Active football player; continuously improving English communication for professional growth.